

Speckle Patterns: Classification and Reconstruction Using Deep Neural Networks

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Abstract

When coherent laser light scatters through a disordered medium, the transmitted wavefront is scrambled into a speckle pattern that appears random to the eye but still encodes information about the original scene. This project investigates whether deep neural networks can decode that information for two complementary tasks: digit classification and image reconstruction. Using the Rice DSP Real Transmission Matrix Dataset across three independently measured scattering media, we model the physical forward process as a complex transmission matrix and train a convolutional neural network (CNN) classifier and a U-Net reconstructor on the resulting intensity patterns. The CNN classifier achieves approximately 95% accuracy at a realistic signal-to-noise ratio of 20 dB, demonstrating that classification from speckle is robust even under significant camera noise. U-Net reconstruction achieves an average peak signal-to-noise ratio (PSNR) of 25.53 dB and structural similarity index (SSIM) of 0.876, only marginally outperforming a plain CNN encoder-decoder (PSNR 25.50 dB, SSIM 0.881). Neither approach significantly generalizes across scattering media: classifiers and reconstructors trained on one transmission matrix converge and have poor generalizability to different medium, a fundamental limitation of investigating speckle patterns across different samples.

1. Introduction

When a laser illuminates a rough scattering medium, the scattered light forms a speckle pattern: a grainy, high-contrast interference pattern that shifts dramatically with even the smallest change in the scattering geometry. To the unaided eye, speckle looks like noise. However, the pattern is entirely deterministic given the input field and the scattering medium. This determinism is the foundation of a growing body of work that uses speckle as a computational imaging channel rather than as a nuisance to be suppressed [1].

Speckle-based imaging has found practical use in laser speckle contrast imaging for blood flow monitoring, tissue perfusion mapping, and eardrum vibration sensing [2]. More recently, researchers have shown that it is possible to use measured transmission matrices (TMs) to focus light through scattering tissue and to reconstruct objects placed behind opaque barriers [3]. Therefore, if a scattering medium can be characterized by its TM, it is possible a neural network learn to invert the forward scattering process directly from intensity measurements.

This project explores that question through two parts. Part 1 addresses classification: given a speckle pattern produced by a known medium, can a CNN predict the digit class of the original image? Part 2 addresses reconstruction: can a U-Net recover a pixel-level estimate of the original digit from the same speckle? We evaluate both tasks across three real TMs from the Rice DSP dataset at multiple noise levels, and we probe the robustness of each approach by testing classifiers and reconstructors on transmission matrices they were not trained on.

2. Related Work

Reconstruction through scattering media has historically been approached with phase retrieval algorithms and iterative matrix inversion. Popoff et al. [3] demonstrated wavefront shaping through multimode fibers using measured TMs, and subsequent work showed that objects could be imaged through opaque diffusers given full TM knowledge. Katz et al. [4] exploited the memory effect of thin diffusers to reconstruct images without measuring the TM at all.

Deep learning approaches to scattering inversion began in earnest with Li et al. [5], who trained a fully connected network to recover images from speckle patterns generated by a thin diffuser. Subsequent work by Lyu et al. [6] and others demonstrated that U-Net architectures could generalize across object classes and modest changes in scattering geometry. However, most prior work used simulated or stationary scattering media.

On the classification side, Turpin et al. [7] showed that CNNs could identify objects from fiber-transmitted speckles at high accuracy, asking the question of whether reconstruction is even necessary for recognition tasks. Our work directly compares these two strategies on the same dataset, using real TMs and controlled noise sweeps, contributing quantitative evidence to this debate.

3. Methods

3.1 Physical Forward Model

We model the scattering medium as a linear map from input field to output field. Given a flattened MNIST digit x treated as a coherent amplitude wavefront, the measured intensity at the detector is:

$$I_{out} = |Tx|^2 + n$$

where T is the complex transmission matrix of dimensions $M \times N$ (M detector pixels, N input pixels), and n is additive Gaussian camera noise. The squared magnitude introduces a nonlinearity element. The noise term n is drawn from a zero-mean Gaussian scaled to achieve a chosen signal-to-noise ratio (SNR) in decibels. We test SNR levels of 40, 20, and 10 dB, plus a noiseless baseline. SNR 20 dB is treated as the primary operating point because it represents realistic camera noise for a scientific CMOS sensor.

3.2 Dataset

All experiments use the Rice DSP Real Transmission Matrix Dataset [8], which provides three independently measured TMs acquired with different spatial light modulator (SLM) configurations:

Table 1. The three transmission matrices from the Rice DSP dataset.

Medium	Input Resolution	Input Pixels (N)	SLM Type
AmpSLM_16x16	16 x 16	256	Amplitude
PhaseSLM_40x40	40 x 40	1,600	Phase
AmpSLM_64x64	64 x 64	4,096	Amplitude

MNIST digits are down sampled to the native input resolution of each TM before forward propagation. The detector array uses a consistent 32 x 32 subsampled grid (1,024 pixels) across all three media, ensuring that the CNN input size is fixed regardless of which TM generated the speckle. Figure 1 shows the resulting speckle patterns for the same digit passed through all three simulated mediums, with and without SNR 20 dB noise.

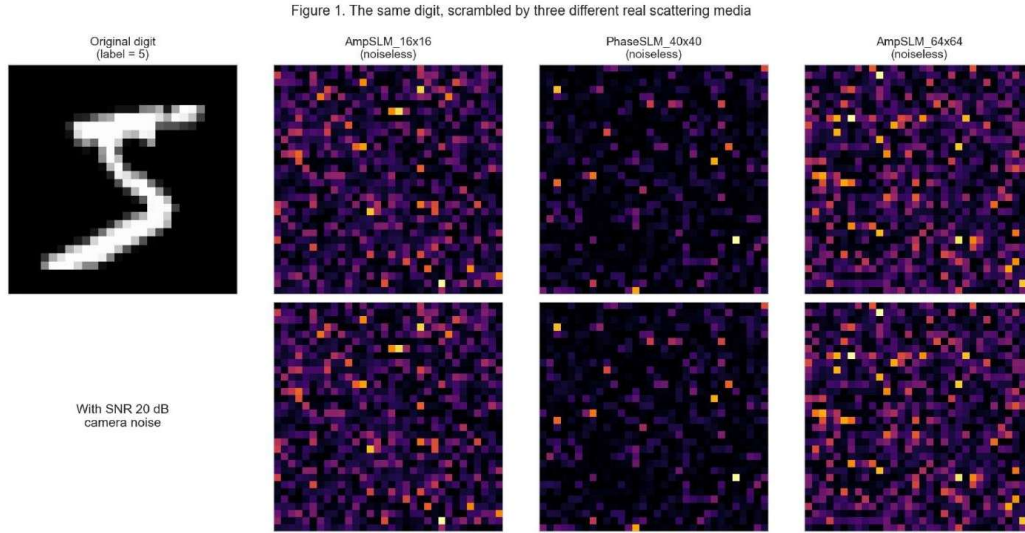


Figure 1. The same MNIST digit (label = 5) scrambled by three real scattering media under noiseless (top) and SNR 20 dB (bottom) conditions.

3.3 Classification Network

The classifier is a standard CNN with two convolutional blocks (Conv2d + MaxPool2d) followed by two fully connected layers and a 10-class SoftMax output. The architecture has 544,522 trainable parameters. The input is always a 32 x 32 speckle intensity image, making the same architecture applicable to all three TMs. Training uses the Adam optimizer with a learning rate of

1e-3, cross-entropy loss, a batch size of 128, and 5 epochs. We use 10,000 training samples and 2,000 test samples per experiment.

3.4 Reconstruction Network

Image reconstruction requires recovering a 16×16 -pixel image from a 32×32 speckle measurement. We compare two architectures: a U-Net with skip connections and a plain CNN encoder-decoder without skip connections. The U-Net uses an asymmetric design: the encoder down samples from 32×32 to 4×4 over three stages with doubling channel depth (32, 64, 128), and the decoder up samples back to 16×16 over two stages. Skip connections concatenate encoder feature maps to their corresponding decoder stages. The plain CNN mirrors the encoder-decoder structure without skip connections. Both models use MSE loss and the Adam optimizer with a learning rate of 1e-3 over 6 epochs, with 15,000 training samples and 2,000 test samples.

Reconstruction quality is measured using PSNR and SSIM [9]. PSNR rewards pixel-level fidelity, while SSIM captures perceptual structural similarity. Both metrics are computed on held-out test samples at each SNR level.

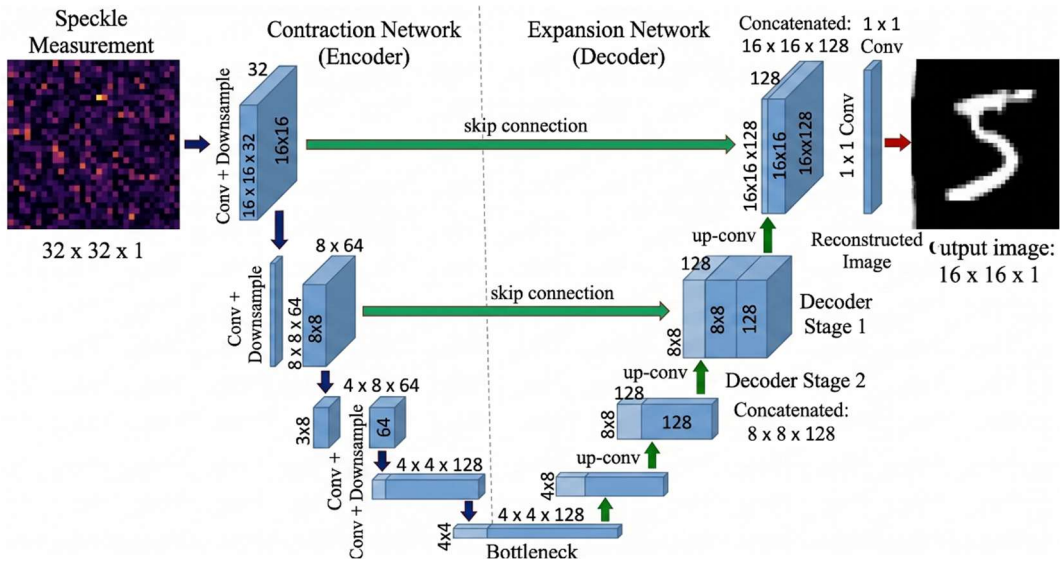


Figure 2. U-Net reconstructor architecture. The encoder down samples from 32×32 to a $4 \times 4 \times 128$ bottleneck over three stages. The decoder (orange) up samples back to 16×16 . Skip connections concatenate encoder feature maps to their decoder counterparts at matching resolutions

3.5 Cross-Transmission-Matrix Experiments

To probe generalization, we train both classifiers and reconstructors on one TM and evaluate them on the other two. For classification, this yields a 3×3 matrix of accuracies where diagonal entries correspond to same-TM evaluation and off-diagonal entries test cross-TM transfer. For reconstruction, we train a U-Net on AmpSLM_16x16 and evaluate it on an unseen randomly generated TM as a negative control, providing a lower bound on performance when the forward model is completely different.

4. Results

4.1 Classification: Noise Robustness

Figure 2 shows classification accuracy on AmpSLM_16x16 as a function of SNR. The CNN achieves 94.4% accuracy in the noiseless condition and remains stable at approximately 95.0-95.1% across the full SNR range from 40 dB to 10 dB. This counterintuitive result, where moderate noise slightly improves accuracy, is likely a regularization effect: noise acts as data augmentation during training, preventing the network from overfitting to exact intensity values. The results demonstrate that speckle-based classification is robust to realistic levels of camera noise.

Table 2. Classification accuracy on AmpSLM_16x16 at varying noise levels.

Condition	Test Accuracy
Noiseless	0.944
SNR 40 dB	0.950
SNR 20 dB	0.951
SNR 10 dB	0.950

4.2 Classification: Cross-TM Generalization

Figure 2 shows the 3 x 3 cross-TM accuracy matrix at SNR 20 dB. The diagonal entries (same-TM evaluation) range from 0.932 to 0.951, confirming that all three media are learnable. However, all six off-diagonal entries fall between 0.082 and 0.136, which is near the 0.10 chance baseline for a balanced 10-class problem. This stark pattern demonstrates that each transmission matrix defines a completely different mapping from input space to speckle space, and the classifier learns an inverse that is entirely tied to the specific medium it was trained on.

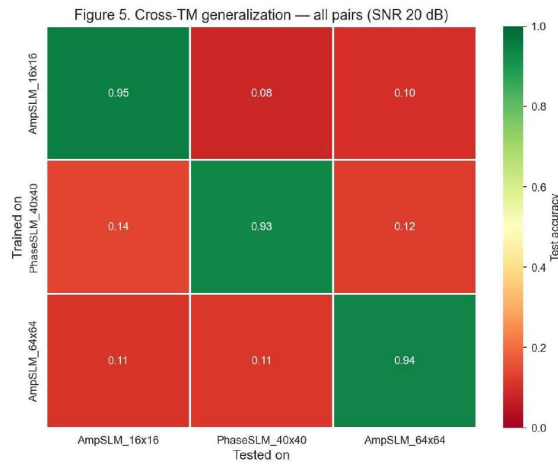


Figure 3. Cross-TM generalization heatmap for classification at SNR 20 dB. Diagonal entries (green) reflect high same-TM accuracy. All off-diagonal entries (red) collapse to near-chance performance, showing medium specificity of speckling.

4.3 Reconstruction: U-Net vs. Plain CNN

Figure 3 shows example reconstructions from the U-Net at SNR 20 dB alongside ground-truth digits. The network recovers the overall shape and stroke structure of most digits, though fine details are blurred. The reconstructions become noisier at 10 dB but remain recognizable for most digit classes.

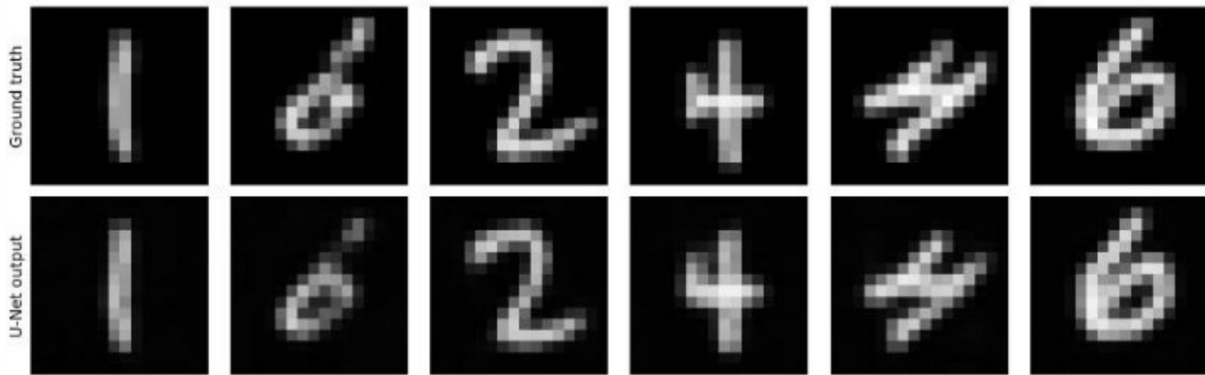


Figure 4. U-Net reconstruction results on the AmpSLM_16x16 medium at SNR 20 dB. Top row: ground-truth digits. Bottom row: U-Net outputs. The network recovers digit identity in most cases despite the globally mixed speckle input.

Figure 4 compares PSNR and SSIM for the U-Net and plain CNN across three SNR levels. Both models perform similarly across the board. The U-Net achieves a marginally higher average PSNR (25.53 dB vs. 25.50 dB), but the plain CNN achieves a slightly higher average SSIM (0.881 vs. 0.876). Neither advantage is practically significant.

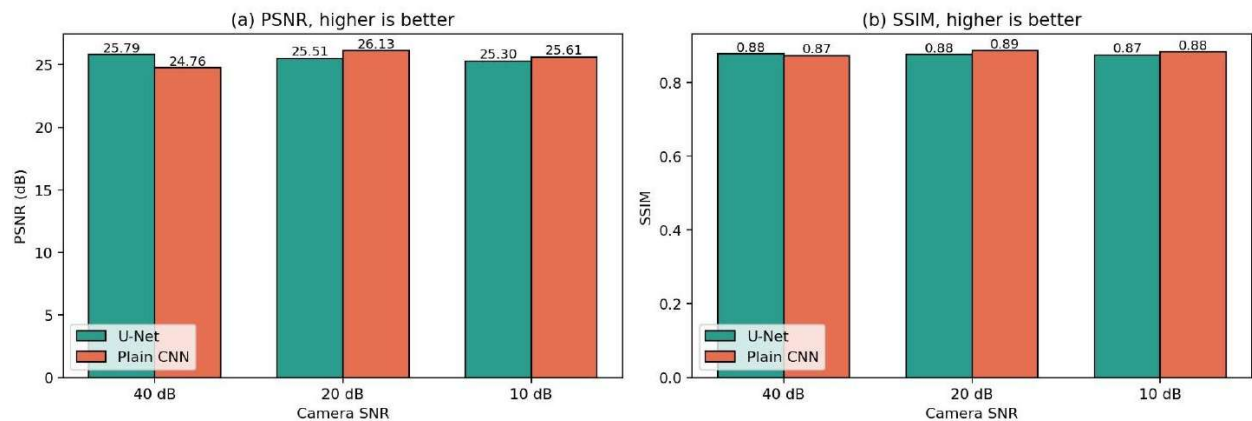


Figure 5. Reconstruction quality (PSNR and SSIM) for U-Net and plain CNN across SNR levels of 40, 20, and 10 dB. The two architectures perform comparably, with differences well within the margin of experimental variance.

Table 3. Average reconstruction performance across all SNR levels (AmpSLM_16x16).

Model	Avg PSNR (dB)	Avg SSIM
U-Net	25.534	0.8759
Plain CNN	25.500	0.8806

4.4 Reconstruction: Cross-TM Generalization

Figure 5 shows the result of applying a U-Net trained on AmpSLM_16x16 to speckle generated by an unseen, randomly generated TM. Performance drops dramatically: PSNR falls from 25.51 dB to 11.76 dB, and SSIM collapses from 0.876 to 0.064. The reconstructed images resemble noise rather than digits, confirming that the U-Net has learned a decoder that is specific to the transmission matrix it was trained on, not a general-purpose speckle inverter.

Figure 4. U-Net trained on real TM, tested on unseen TM

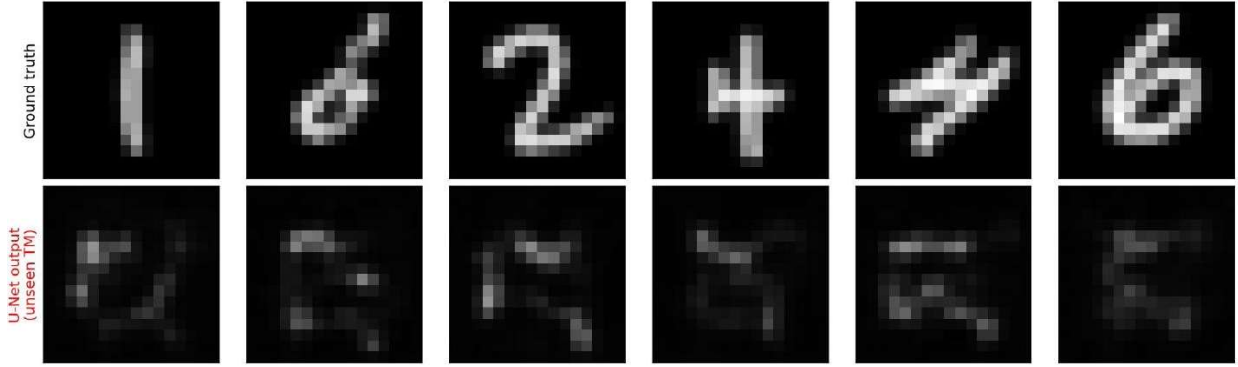


Figure 6. Cross-TM reconstruction failure. Top row: ground-truth digits. Bottom row: U-Net outputs when tested on an unseen transmission matrix. PSNR drops from 25.51 dB to 11.76 dB and SSIM collapses from 0.876 to 0.064, demonstrating that the learned inverse is entirely medium-specific.

5. Discussion

5.1 Why Skip Connections Provide Limited Benefit

The near-equivalence of U-Net and plain CNN reconstruction performance is unexpected given U-Net's strong track record in biomedical image segmentation. The explanation lies in speckle the transmission matrix T acts as a globally mixing linear operator, so each output pixel depends on all N input pixels. This destroys the local spatial structure that skip connections are designed to preserve. In a standard segmentation task, a skip connection from an early encoder layer carries spatial edge information that helps the decoder place boundaries accurately. In the speckle case, early encoder feature maps contain scrambled global mixtures of the input, providing no spatial locality that the decoder can leverage.

5.2 The Cross-TM Generalization Problem

There is a complete absence of cross-TM generalization. This is not a failure of the specific architecture used but a fundamental consequence of the measurement model. The intensity measurement ($|Tx|^2$) is a different nonlinear function of x for each T . A network that learns to invert one T cannot invert a different T without retraining, because the two tasks are unrelated.

This has direct implications for practical systems. A speckle-based sensor deployed in a clinical or industrial setting will encounter changes in the scattering medium due to temperature drift, mechanical perturbation, or biological variation. Each change effectively replaces T with a dramatically new matrix T' , and a pre-trained network will fail catastrophically. Potential mitigations include online TM recalibration, domain adaptation techniques, or physics-informed architectures that explicitly incorporate TM knowledge as a learnable parameter.

5.4 Limitations and Future Work

Several limitations of this study merit acknowledgment. First, all experiments use down sampled MNIST digits, which have simpler structure than real biomedical images. Second, training on 10,000-15,000 samples may underutilize the expressive capacity of the networks; larger datasets might reduce the U-Net vs. CNN gap.

Future work could explore physics-informed architectures that take the TM as an explicit input, enabling a single network to operate across multiple media. Alternatively, meta-learning approaches could be used to learn a rapid adaptation strategy that fine-tunes to a new TM using only a small calibration set. On the reconstruction side, incorporating an adversarial discriminator might improve visual quality.

6. Conclusion

This project demonstrates that speckle patterns produced by real scattering media are not random noise but structured, learnable representations of the original scene. A CNN classifier achieves approximately 95% accuracy from speckles at realistic noise levels, and a U-Net reconstructor recovers recognizable digit images. The two approaches are complementary: classification is faster, more parameter-efficient, and more noise-robust, while reconstruction is interpretable and applicable to open-ended downstream tasks.

However, both approaches share a fundamental limitation: they learn inverses that are specific to the transmission matrix used during training. Cross-TM evaluation reveals near-chance classification and noise-like reconstruction when the scattering medium changes. This medium specificity is not an artifact of the chosen architecture but a mathematical consequence of the scattering forward model. Addressing this limitation is the central open problem for deploying speckle-based deep learning in dynamic real-world environments.

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